AGRICULTURAL EQUIPMENT & FARM MACHINERY

Illustrated field-use catalogue — with a focus on Andhra Pradesh, India



From hand tools and mini equipment to tractors, harvesters, irrigation systems and specialised machines.

Prepared as a practical reference for farmers, students, field teams, agricultural projects, equipment buyers and Custom Hiring Centres (CHCs).

Important scope note: farming has many thousands of models and locally modified tools. This guide is a broad practical catalogue of major equipment families and common examples, not a claim that every brand, model or locally fabricated implement is listed.

How to use this guide

Equipment is grouped by the job it performs in the field. A machine can be available in manual, animal-drawn, engine-powered, battery-powered, tractor-mounted or self-propelled versions. Choose by the job and the field conditions first, then choose the machine size.

Andhra Pradesh crop context

The examples emphasise paddy, maize, groundnut, cotton, chilli, pulses, sugarcane, oilseeds, vegetables, mango, banana, coconut, oil palm and other horticultural crops. Actual crop mix varies by district, season, irrigation access and farm size.

Size categories used

Equipment / machine	Size / type	Main field use	Especially relevant for
Hand tools	Hand-operated	Low-cost tasks, small plots, repair work and precision work	All crops
Mini / portable	Manual, battery or small engine	Small fields, narrow rows, orchards and labour-saving tasks	Vegetables, chilli, orchards, small paddy plots
Walk-behind / power tiller	Small engine, 2-wheel	Tillage, weeding, puddling and compatible attachments	Small to medium plots
Tractor-operated	Mounted, trailed or PTO-driven	Higher field capacity and multiple implement options	Medium to large fields
Self-propelled / specialised	Dedicated machine	High-volume or crop-specific operations	Large paddy, maize, sugarcane, commercial farms

Field-safety note: guards, shields, PTO covers, brakes, emergency stops and proper PPE matter. Keep people clear of rotating shafts, belts, blades and moving harvest mechanisms. Use pesticides only according to label directions and local guidance.

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1. Hand tools and small equipment

Basic tools are often the most affordable and flexible option. Many remain essential even on mechanised farms for field edges, repairs, spot weeding, harvesting and horticultural work.



Equipment / machine	Size / type	Main field use	Especially relevant for
Hand hoe / grubbing hoe	Hand tool	Loosening soil, removing weeds and shaping small beds	Vegetables, chilli, groundnut
Spade / shovel	Hand tool	Digging, moving soil, compost and manure	All crops
Pickaxe / mattock	Hand tool	Breaking hard soil and digging pits	Orchards, bunds, hard ground
Sickle	Hand tool	Cutting grass, fodder and crop stalks	Paddy, fodder, field edges
Axe / billhook / machete	Hand tool	Cutting woody growth and trimming	Orchards, plantations
Crowbar / digging bar	Hand tool	Levering stones and opening compacted spots	Field preparation
Hand weeder / push-pull weeder	Manual	Removing weeds between crop rows	Vegetables, row crops
Wheel hoe	Manual wheeled	Fast inter-row weeding on prepared beds	Vegetables, pulses
Manual seed dibbler	Manual	Placing seed at regular spacing and depth	Maize, pulses, vegetables
Pruning secateurs / loppers	Hand tool	Pruning shoots and branches	Mango, citrus, grapes, plantations
Fruit picker / mango harvester	Pole-mounted	Picking fruit with less bruising or climbing	Mango and fruit orchards
Harvesting knife / bunch cutter	Hand tool	Cutting banana bunches and other produce	Banana, horticulture
Wheelbarrow / hand trolley	Manual transport	Moving inputs, produce and tools within a farm	All crops
Measuring tape, rope and field markers	Manual	Row layout, spacing and plot measurement	Planting and irrigation layout
Manual knapsack sprayer	Hand-pumped	Small-scale spot spraying where label-approved	Small plots and orchards
Manual seed cleaner / sieves	Manual	Small-batch cleaning and grading	Seeds, pulses, grain

2. Land preparation, tillage and levelling

Select implements according to soil texture, moisture, previous crop, residue and the depth of tillage required. Repeated intensive tillage is not always necessary.



Equipment / machine	Size / type	Main field use	Especially relevant for
Country plough / indigenous plough	Animal-drawn or small tractor	Basic soil opening and furrow making	Small fields, traditional systems
Mouldboard plough	Tractor-mounted	Inverting soil and burying weeds/residue	Fields requiring primary tillage
Disc plough	Tractor-mounted	Cutting and turning tougher or residue-rich soil	Hard, sticky or trashy soils
Chisel plough	Tractor-mounted	Cracking soil with less inversion	Compaction management
Subsoiler / ripper	Heavy tractor-mounted	Breaking a diagnosed compacted layer at depth	Compacted fields; use only when needed
Cultivator / tine cultivator	Tractor-mounted	Secondary tillage and weed uprooting	Maize, pulses, cotton, dryland crops
Disc harrow	Tractor-mounted	Breaking clods and mixing surface residues	Seedbed preparation
Spike-tooth / peg-tooth harrow	Tractor or animal-drawn	Light clod breaking and levelling	Seedbed finishing
Rotavator / rotary tiller	PTO-driven or mini	Pulverising soil and mixing residue	Many crops; avoid unsuitable wet or compacted conditions
Power tiller	Walk-behind	Tillage, puddling and attachment-based field work	Small and fragmented plots
Puddler	Tractor or power-tiller attachment	Preparing wet soil for transplanted paddy	Paddy
Laser land leveler	Tractor-mounted with laser system	Precise land levelling and more uniform irrigation	Suitable irrigated fields
Land plane / blade leveler	Tractor or animal-drawn	General levelling and field shaping	Most field crops
Bund former / ridger	Tractor, power tiller or manual	Making ridges, furrows and field bunds	Groundnut, maize, vegetables, sugarcane
Bed former / raised-bed maker	Tractor-mounted	Creating raised beds and furrows	Vegetables, chilli, maize
Rotary ditcher / trencher	Tractor-mounted or dedicated	Opening drainage or irrigation channels	Fields needing planned drainage
Stone picker / land clearing tools	Manual or mechanical	Removing stones and obstacles	Selected fields

3. Sowing, planting and transplanting

Correct seed rate, seed depth, spacing and calibration are as important as machine choice. Confirm the crop-specific setting before planting.



Equipment / machine	Size / type	Main field use	Especially relevant for
Broadcasting tray / hand broadcaster	Manual	Scattering seed or granular inputs	Paddy, fodder and selected crops
Seed-cum-fertilizer drill	Tractor-drawn	Sowing seed and placing fertilizer in rows	Maize, pulses, oilseeds
Zero-till drill	Tractor-mounted	Direct drilling with minimal soil disturbance	Suitable cereals and rotations
Multi-crop planter	Tractor-mounted	Metered row planting with adjustable plates/rollers	Maize, groundnut, pulses, cotton depending on model
Precision planter	Tractor-mounted	Accurate seed spacing and depth	Commercial row crops
Pneumatic planter	Tractor-mounted	Air-assisted precision seed placement	Larger precision-seeding operations
Groundnut planter / seed drill	Tractor or power tiller	Planting kernels at set spacing and depth	Groundnut
Maize planter	Tractor-mounted	Row sowing and often fertilizer placement	Maize
Cotton planter	Tractor-mounted	Sowing cotton seed at regular spacing	Cotton
Paddy drum seeder	Manual push type	Direct seeding in prepared wet fields	Paddy
Wet direct-seeded rice drill	Tractor/power tiller type	Sowing rice in rows in suitable prepared fields	Paddy
Rice transplanter	Walk-behind or riding	Transplanting nursery-raised rice seedlings	Paddy
Vegetable transplanter	Manual, semi-auto or tractor type	Transplanting vegetable seedlings at uniform spacing	Chilli, tomato, cabbage and other vegetables
Potato / tuber planter	Tractor-mounted	Planting tubers and forming rows	Where tuber crops are grown
Sugarcane planter	Tractor-mounted	Opening furrow, placing setts and covering them	Sugarcane
Nursery tray seeder	Bench or powered	Filling trays and sowing seeds uniformly	Vegetable and horticulture nurseries
Seed treatment drum	Small powered/manual	Coating seed with approved treatment uniformly	Seeds where recommended
Fertilizer applicator / side dresser	Manual or tractor-mounted	Placing fertilizer near the crop row	Row crops

4. Weeding, interculture and crop maintenance

Early weed control can be critical. Match the implement to row spacing, crop stage, soil moisture and the risk of root damage.



Equipment / machine	Size / type	Main field use	Especially relevant for
Hand hoe / wheel hoe	Manual	Spot and inter-row weeding	Vegetables, chilli, pulses
Cono weeder	Push type	Weeding between rice rows in suitable spacing	Transplanted paddy
Rotary paddy weeder	Manual or powered	Inter-row weed control in rice	Paddy
Power weeder	Walk-behind / mini engine	Faster inter-row cultivation	Vegetables, chilli, orchards, row crops
Power tiller with weeder attachment	Walk-behind	Weeding and shallow cultivation	Small fields
Inter-row cultivator	Tractor-mounted	Shallow cultivation between rows	Maize, cotton, groundnut
Sweep cultivator / duck-foot sweep	Tractor-mounted	Cutting weeds beneath the soil surface	Row crops
Rotary hoe	Tractor-mounted	Breaking crust and uprooting small weeds	Selected row crops
Mechanical finger weeder	Tractor or tool-carrier	Weeding close to plants in rows	Selected vegetables and row crops
Mulcher / flail mower	Tractor-mounted	Chopping weeds and crop residues	Orchards and residue management
Brush cutter	Portable engine/battery	Cutting grass and weeds around field boundaries	Orchards, bunds, field edges
Hedge trimmer / pruning tool	Powered or manual	Trimming hedges and shaping plants	Nurseries, orchards, plantations
Plastic mulch laying machine	Tractor-mounted	Laying film and sometimes drip line	Chilli and vegetable cultivation
Earthing-up attachment	Tractor/power tiller	Drawing soil around plant bases	Groundnut, potato and other suitable crops
Fertilizer broadcaster	Manual, tractor or PTO-driven	Spreading granular fertilizer uniformly	Open fields; calibrate carefully
Manure spreader	Tractor-trailer type	Distributing compost or farmyard manure	Field fertility management

5. Spraying, dusting and crop protection

Choose spray equipment based on the crop canopy, target pest/disease, label directions, droplet requirements and field size. Avoid drift to people, livestock, water and neighbouring crops.

Equipment / machine	Size / type	Main field use	Especially relevant for
Hand compression sprayer	Manual tank	Small spot applications	Nurseries and small plots
Knapsack sprayer	Manual pump	Small and medium plot spraying	Chilli, vegetables, pulses
Battery sprayer	Battery-powered	Lower-pumping-effort spraying	Small farms and horticulture
Engine-powered knapsack sprayer	Portable engine	Higher-output spraying or misting	Selected crops and orchards
Mist blower / duster	Portable engine	Air-assisted application where label and crop permit	Orchards and selected crops
Power sprayer with hose and lance	Motor/pump type	Pressurised spray for multiple rows or trees	Orchards, chilli, plantations
Tractor-mounted boom sprayer	Tractor-mounted	Uniform broad-acre spray across rows	Field crops
Tractor-mounted air-blast sprayer	Tractor-mounted	Canopy penetration in orchards	Fruit orchards
High-clearance sprayer	Self-propelled	Spraying over taller crop canopies	Large commercial fields
Drone sprayer	UAV, crop-approved setup	Targeted aerial application by trained operators	Paddy, maize, chilli and other suitable crops
Granular applicator	Manual or tractor-mounted	Placing permitted granular crop inputs	Crop-specific use
Bird scarer / acoustic deterrent	Manual or powered	Deterring birds where permitted and effective	Seed and fruit protection
Insect-proof nets / row covers	Passive equipment	Physical exclusion of selected pests	Nurseries and vegetables
Pheromone / sticky trap supports	Manual field devices	Monitoring and selected integrated pest management uses	Crop-specific monitoring

Safety: never spray against the wind or without suitable PPE. Do not mix products unless labels and trained local advice allow it. Drone operation may require compliance with current aviation, pesticide and state rules.

6. Irrigation, drainage and water management

Water systems include pumps, pipes, field distribution, filtration and drainage. Pump sizing depends on water source, lift, required flow, pipe friction and energy supply.



Equipment / machine	Size / type	Main field use	Especially relevant for
Centrifugal pump set	Electric or engine-driven	Lifting and transferring surface water	Canals, ponds, tanks
Submersible pump	Electric	Pumping from borewells and deep sources	Irrigated farms
Diesel pump set	Portable engine-driven	Water pumping where electricity is unavailable	Fields and backup use
Solar pump set	Solar-powered	Daytime irrigation with lower grid/diesel dependence	Suitable water sources and field layouts
Pump starter, control panel and protection	Electrical	Safe motor control and protection	Electric pump systems
Suction hose, delivery hose and foot valve	Pipes/fittings	Connecting and protecting pump intake/delivery	Pump installations
HDPE/PVC pipes and fittings	Main/sub-main lines	Moving water across the farm	All irrigated crops
Drip irrigation system	Emitters and pipes	Delivering water near the root zone	Chilli, vegetables, orchards, plantations
Drip filter and fertigation unit	Filter, venturi/doser	Filtering water and applying compatible nutrients	Drip-irrigated crops
Micro-sprinkler system	Low-flow sprinklers	Localised irrigation and nursery/orchard watering	Horticulture
Sprinkler / rain-gun system	Portable or fixed	Applying water over a wider area	Suitable field crops
Portable sprinkler set	Pipe and sprinkler	Flexible irrigation on small or medium plots	Vegetables, pulses, fodder
Furrow irrigation tools	Ridger, gates, siphon tubes	Directing water through furrows	Row crops
Water meter / pressure gauge	Monitoring instruments	Checking flow and pressure	All pressurised systems
Soil-moisture sensor	Probe-based	Guiding irrigation timing	Drip and precision irrigation
Field drainage pump	Portable or fixed	Removing excess water from low areas	Waterlogged fields
Open-channel / field ditcher	Tractor-mounted or manual	Making planned drainage channels	Low-lying fields
Land leveller and field bund tools	Tractor/manual	Improving distribution and containing water	Paddy and irrigated fields
Rain gauge and farm weather station	Manual/electronic	Tracking rainfall and local conditions	All crops

7. Harvesting and crop-specific machines

Harvesting machinery is crop-specific. Check field access, soil bearing strength, crop maturity, grain moisture, machine width and transport arrangements before booking a machine.



Equipment / machine	Size / type	Main field use	Especially relevant for
Sickle / harvesting knife	Hand tool	Manual cutting of stalks and grasses	All crops at small scale
Paddy reaper	Walk-behind or self-propelled	Cutting standing rice and laying it in rows	Paddy
Reaper-binder	Self-propelled or tractor-mounted	Cutting and tying crop bundles	Suitable cereals
Combine harvester	Self-propelled or tractor-driven	Harvesting, threshing and cleaning in one pass	Paddy, wheat and other compatible crops
Mini combine harvester	Compact self-propelled	Combining in smaller fields where access permits	Paddy and compatible crops
Maize picker / maize harvester	Tractor-mounted or self-propelled	Picking ears or harvesting maize depending on design	Maize
Maize sheller	Stationary or mobile	Removing kernels from cobs	Maize
Groundnut digger / inverter	Tractor-mounted	Lifting plants and loosening pods from soil	Groundnut
Groundnut pod stripper / picker	Powered or tractor-operated	Separating pods from plants	Groundnut
Cotton picker	Hand-held or large machine	Picking open cotton bolls	Cotton; machine type depends on scale
Sugarcane cutter / harvester	Manual tools or large self-propelled	Cutting stalks and sometimes topping/chopping	Sugarcane
Sugarcane loader	Tractor-mounted or dedicated	Loading harvested cane	Sugarcane
Forage mower	Manual, walk-behind or tractor-mounted	Cutting fodder grass and forage crops	Fodder
Forage harvester / chopper	Tractor-mounted or self-propelled	Chopping green fodder or whole-crop silage	Dairy and livestock farms
Baler	Tractor PTO-driven	Compressing straw or hay into bales	Fodder and residue handling
Mango harvester / fruit picker	Pole or platform system	Harvesting fruit with reduced damage	Mango and orchards
Coconut climbing device / harvester tools	Manual safety device or powered tools	Supporting safe coconut harvesting	Coconut plantations
Oil palm bunch harvesting tools	Long-handled specialised tools	Cutting ripe fresh fruit bunches	Oil palm
Banana bunch cutter and trolley	Hand tool and transport	Cutting and moving bunches carefully	Banana

Equipment / machine	Size / type	Main field use	Especially relevant for
Chilli picking aids	Manual or crop-specific tools	Supporting harvest and handling	Chilli
Potato / tuber digger	Tractor-mounted	Lifting tubers from soil	Where tuber crops are grown

8. Threshing, cleaning, drying and storage

Post-harvest equipment reduces handling losses and helps maintain quality. Keep equipment clean and set to the crop and grain condition.

Equipment / machine	Size / type	Main field use	Especially relevant for
Paddy thresher	Manual, engine or PTO-driven	Separating rice grain from harvested panicles	Paddy
Multi-crop thresher	Stationary powered	Threshing compatible cereals and pulses with adjustments	Paddy, maize, pulses where compatible
Maize sheller	Hand, motor or PTO-driven	Removing kernels from cobs	Maize
Groundnut decorticator	Powered or manual	Separating kernels from pods	Groundnut
Pulse thresher / dal pre-cleaner	Powered	Threshing or cleaning pulse crops	Pulses
Winnower	Manual or powered fan	Separating chaff and light impurities	Grains and pulses
Grain cleaner / grader	Sieve and air system	Removing impurities and grading by size	Seeds, grains, pulses
Seed grader	Precision screens/air	Sorting seed lots for quality	Seed production
Grain dryer	Batch or continuous-flow	Reducing grain moisture under controlled conditions	Paddy, maize and grains
Solar dryer	Solar heat	Drying crops with protection from dust and rain	Chilli, fruits, vegetables
Moisture meter	Handheld	Checking grain or seed moisture	Before storage and sale
Rice mill / mini rice mill	Powered processing unit	Husking and milling paddy	Paddy-growing regions
Dal mill / pulse mill	Powered processing unit	Dehusking and splitting pulses	Pulses
Oil expeller	Small or commercial	Extracting oil from suitable oilseeds	Groundnut and other oilseeds
Chilli cleaner / destoner	Powered	Removing dust, stones and foreign material	Chilli
Chilli stem cutter / grader	Powered or semi-automatic	Removing stems and grading dried chilli	Chilli
Turmeric washer / polisher	Powered	Cleaning and polishing turmeric rhizomes	Turmeric
Turmeric boiler / dryer	Batch equipment	Curing and drying turmeric	Turmeric
Fruit washer / grader / packer	Manual or powered line	Cleaning, sizing and packing produce	Mango and horticulture
Weighing scale and bag stitcher	Manual or powered	Weighing, bagging and closing produce	Grain and produce
Hermetic bags / grain bins / silos	Storage systems	Protecting dry grain from moisture and pests	Grain and seed storage
Crates, pallets and bulk bags	Handling equipment	Reducing bruising and improving loading	Fresh produce and grain

9. Residue management and soil health

Residue management can protect soil, recycle nutrients and reduce burning. Choose the method according to the next crop, residue volume, moisture and local disease/pest risks.



Equipment / machine	Size / type	Main field use	Especially relevant for
Straw chopper / straw shredder	Tractor-mounted or combine attachment	Chopping standing or harvested residue	Paddy and cereals
Mulcher / flail shredder	Tractor-mounted	Chopping weeds and crop residues	Orchards, field residues
Rotary slasher	Tractor-mounted	Cutting grass and coarse vegetation	Field edges and fallow land
Baler	PTO-driven	Compressing straw into transportable bales	Paddy straw, fodder
Rake / hay rake	Manual or tractor-mounted	Gathering cut grass or straw into rows	Fodder and residue
Happy Seeder / residue-retaining drill	Tractor-mounted	Sowing into suitable retained cereal residue	Suitable cereal rotations
Zero-till residue drill	Tractor-mounted	Direct seeding through surface residue	Compatible crops and systems
Compost shredder	Small powered	Reducing organic material size before composting	Farm organic waste
Compost turner	Manual, tractor or dedicated	Turning compost piles for aeration	Compost production
Vermicompost sieves and beds	Manual/small equipment	Handling and screening vermicompost	Farm nutrient management
Manure spreader	Tractor-trailer	Applying organic manure more evenly	Field crops
Biochar kiln / biomass unit	Small or specialised	Processing biomass under controlled conditions	Only where locally suitable and safely managed
Soil sampling auger and kit	Manual	Collecting representative soil samples	All crops
Portable soil testing kit	Field kit	Indicative field measurements; lab tests may still be needed	Soil fertility management

10. Farm transport, loading and field logistics

Transport equipment is often overlooked, but it directly affects harvest speed, produce damage, labour needs and turnaround time.

Equipment / machine	Size / type	Main field use	Especially relevant for
Tractor trailer	Tractor-drawn	Moving inputs, harvest and farm materials	All crops
Power-tiller trailer	Small powered	Transporting produce on small farms	Small plots and orchards
Mini farm tipper	Powered vehicle or tractor trailer	Moving soil, manure and produce with tipping unload	Mixed farms
Farm trolley / hand cart	Manual	Moving crates and inputs within the field	All crops
Three-wheeler cargo vehicle	Small motor vehicle	Moving produce on narrow rural roads	Small farms and market links
Conveyor belt	Powered	Moving grain, bags or produce over short distances	Threshing and packing areas
Bag lifter / sack trolley	Manual or powered	Reducing lifting effort for bags	Grain and input handling
Pallet jack / stacker	Manual or powered	Moving palletised goods in stores and packhouses	Larger farms and aggregation points
Front-end loader	Tractor-mounted or dedicated	Loading bulk materials and manure	Large farms and yards
Forklift	Powered	Handling pallets and bulk bags	Packhouses and warehouses
Field crates and harvest bins	Reusable containers	Protecting fresh produce during movement	Fruit and vegetables
Platform scale / weighbridge	Manual/electronic	Weighing crop loads and farm inputs	Aggregation and sale points

11. Horticulture, orchards and plantation equipment

Orchards and plantations often need compact machines, careful pruning and harvesting tools, sprayers suited to canopy shape, and transport that avoids bruising.

Equipment / machine	Size / type	Main field use	Especially relevant for
Mini tractor / narrow-track tractor	Compact tractor	Working in narrow rows and smaller holdings	Orchards, plantations, vegetables
Orchard power sprayer	Portable or tractor-mounted	Spraying tree canopies with controlled coverage	Mango, citrus and other orchards
Air-blast sprayer	Tractor-mounted	Air-assisted canopy spray	Commercial orchards
Pruning saw, secateurs and loppers	Manual/battery	Pruning and canopy management	Mango, citrus, grape, plantations
Pole pruner / pole fruit picker	Long-handled	Reaching branches and fruit from the ground	Mango and fruit crops
Orchard mower / inter-row mower	Walk-behind or tractor	Managing grass between tree rows	Orchards and plantations
Brush cutter	Portable	Managing field edges and undergrowth	Orchards and plantations
Drip line laying machine	Tractor-mounted	Installing drip pipe during bed preparation	Chilli, vegetables, orchards
Plastic mulch layer	Tractor-mounted	Laying mulch film and sometimes drip line	Chilli and vegetables
Nursery pot/tray filling machine	Manual or powered	Filling nursery containers consistently	Horticulture nurseries
Seedling tray seeder	Semi-automatic	Sowing small seeds into plug trays	Vegetable nurseries
Fruit grading table / roller grader	Manual or powered	Sorting produce by size and quality	Mango, citrus and vegetables
Coconut dehusker	Manual or powered	Removing coconut husk	Coconut
Coconut climbing safety device	Manual safety equipment	Supporting climbing and harvesting	Coconut; use trained operators
Oil palm harvesting chisel / sickle	Long-handled specialised	Cutting fresh fruit bunches	Oil palm
Banana stem chopper	Powered or manual	Managing harvested stems and residues	Banana
Cashew nut shelling / grading equipment	Processing equipment	Post-harvest handling and value addition	Cashew-growing areas
Shade nets, trellis posts and tying tools	Field infrastructure	Supporting plants and managing protected cultivation	Vegetables, grapes and nurseries

12. Farm power, tractor types and attachments

The tractor is a power source for many implements, but horsepower alone does not guarantee compatibility. Check implement working width, weight, hitch category, PTO speed, hydraulic capacity, tyre/track setup and local terrain.

Equipment / machine	Size / type	Main field use	Especially relevant for
Compact / mini tractor	Low horsepower	Light tillage, spraying, transport and orchard work	Small farms and horticulture
Utility tractor	Medium horsepower	General tillage, sowing, trailers and PTO work	Mixed crop farms
High-horsepower tractor	Higher horsepower	Heavy tillage, large implements and demanding operations	Large fields and heavy soils
Two-wheel power tiller	Walk-behind	Tillage and attachment-driven field work	Small and fragmented holdings
Crawler / track-type tractor	Tracks	Traction in selected soft-soil conditions	Specialised operations
Front-end loader attachment	Hydraulic attachment	Loading manure, soil and bulk materials	Farm yards
Backhoe / trenching attachment	Tractor or dedicated machine	Ditches, pits and drainage work	Infrastructure and field development
Rotavator attachment	PTO-driven	Rotary tillage	Seedbed preparation
Cultivator / harrow attachment	Three-point linkage	Tillage and weed control	Row and field crops
Ridger / bed former	Mounted or trailed	Making ridges, furrows and raised beds	Groundnut, maize, vegetables
Seed drill / planter	Mounted or trailed	Row sowing	Crop-specific
Sprayer / fertilizer spreader	Mounted or trailed	Applying inputs uniformly	Field crops
Trailer / tipper	Drawbar	Farm transport	All crops
PTO shaft and guards	Compatible rated shaft	Transferring tractor power to implements	PTO-driven machinery
Hydraulic hoses, couplers and ballast	Tractor-specific	Supporting safe attachment operation and stability	As specified by manufacturer

Never improvise PTO shaft lengths or remove guards. A dealer or trained mechanic should verify hitching, shaft speed, implement weight and tractor stability before first use.

13. Sensors, drones and precision-farming equipment

Digital equipment can help with monitoring and timing, but it should support—not replace—field scouting, crop knowledge and local agricultural advice.

Equipment / machine	Size / type	Main field use	Especially relevant for
Agricultural drone for mapping	Multirotor/fixed-wing as approved	Mapping field boundaries, crop variability and damage	Medium and large farms, service providers
Agricultural spray drone	Crop-specific payload	Applying permitted products under trained operation	Selected crops and service models
GNSS/GPS field mapper	Handheld or mounted	Recording field boundaries and sampling locations	Farm planning
Tractor guidance / auto-steer	GNSS guidance system	Improving pass alignment and reducing overlap	Larger mechanised fields
Soil-moisture sensor	Probe-based	Helping schedule irrigation	Drip-irrigated crops
Weather station	Field sensor station	Recording rainfall, temperature, humidity and wind	All crops
Leaf wetness / canopy sensors	Electronic	Monitoring conditions relevant to some disease risks	Horticulture and field crops
Flow meter and pressure sensor	Irrigation instruments	Checking irrigation system performance	Drip and sprinkler systems
Crop camera / scouting phone tools	Portable digital	Documenting crop health and pest symptoms	All crops
Electronic weighing scale	Digital	Recording harvest and input weights	Farm and aggregation use
Smart irrigation controller	Timer/sensor controlled	Automating irrigation valves or pumps	Pressurised irrigation
Portable chlorophyll / crop sensor	Optical	Indicative crop status measurements	Where supported by local recommendations
Electronic grain moisture meter	Handheld	Checking drying and storage readiness	Grains and seed
Solar fencing controller	Solar/electric	Managing wildlife intrusion where lawful and safe	Farm boundaries

Drone and electrical systems must be operated by trained people and comply with current Indian regulations, pesticide labels, aviation rules and local permissions. Never use improvised electrified fencing.

14. Equipment selection by crop and farm size

This is a starting shortlist, not a fixed prescription. Local field conditions and custom-hiring availability can change the best choice.

Equipment / machine	Size / type	Main field use	Especially relevant for
Paddy / rice	Crop shortlist	Power tiller or tractor, puddler, drum seeder or transplanter, cono weeder, reaper/combine, thresher, dryer, moisture meter	Wet-field suitability, access to bunds, field size and harvesting schedule
Maize	Crop shortlist	Seed-cum-fertilizer drill or planter, cultivator, sprayer, maize harvester/picker, sheller, dryer	Row spacing, stalk handling, grain moisture and market quality
Groundnut	Crop shortlist	Planter, ridger, interculture tools, digger/inverter, pod stripper, cleaner	Soil texture, pod loss risk and digging depth
Cotton	Crop shortlist	Precision planter, inter-row cultivator, sprayer, cotton picking tools/machine, trailer	Row spacing, boll maturity and farm scale
Chilli	Crop shortlist	Nursery tray equipment, transplanter or manual planting aids, drip/fertigation, power weeder, sprayer, picking and drying equipment	Plant spacing, canopy, residue and drying quality
Pulses	Crop shortlist	Seed drill/planter, cultivator, sprayer, reaper, pulse thresher, cleaner/grader	Seed size, shattering risk and crop-specific threshing settings
Sugarcane	Crop shortlist	Furrow opener/ridger, sett planter, interculture tools, sprayer, cane cutter/harvester, loader and trailer	Row width, field access, harvest volume and transport
Vegetables	Crop shortlist	Bed former, precision seeder, seedling tray seeder, transplanter, drip/mulch layer, power weeder, sprayer, crates and grader	Crop spacing, small field access and delicate produce handling
Mango / fruit orchards	Crop shortlist	Mini tractor, mower, pruning tools, orchard sprayer, fruit picker, crates, grading table	Tree spacing, canopy, fruit handling and slope
Coconut / oil palm	Crop shortlist	Plantation tools, safe climbing/harvesting equipment, sprayer, trolley, chopper, irrigation system	Worker safety, tree height, terrain and plantation spacing
Fodder / dairy-linked crops	Crop shortlist	Mower, forage chopper, rake, baler, trailer and silage equipment	Harvest frequency, storage plan and livestock demand

Small farm vs. large farm

Equipment / machine	Size / type	Main field use	Especially relevant for
Small / fragmented holding	Crop shortlist	Hand tools, battery sprayer, power weeder, power tiller, small pump, manual or compact planting tools	Prioritise versatile machines, easy transport and rental access
Medium holding	Crop shortlist	Utility tractor with shared implements, seed drill, rotavator where suitable, trailer, sprayer and threshing access	Compare ownership with seasonal hiring
Large holding / service provider	Crop shortlist	Higher-capacity tractor, precision planter, combine, baler, large sprayer, grain handling and drying systems	Use utilisation, operator availability and maintenance costs to justify investment
Custom Hiring Centre (CHC)	Crop shortlist	A mix of commonly requested implements and crop-specific machines	Select equipment based on local crops, peak-season demand, repair support and booking records

15. Maintenance, safety and buying checklist

Before purchase or rental

• Define the operation: tillage, planting, weeding, spraying, harvesting, transport or post-harvest processing.

• Confirm the machine works with the crop, row spacing, soil moisture, field size, bunds and access roads.

• Check rated capacity, working width, engine/tractor power requirement, fuel or battery needs and expected field capacity.

• Ask for a live demonstration in a field similar to yours; check losses, quality, finish and operator comfort.

• Confirm spare parts, service location, warranty, consumables, operator training and repair turnaround.

• Compare full cost per acre or hour: rental, fuel, operator, transport, maintenance and downtime—not only purchase price.

• For shared equipment, agree on booking, transport, cleaning, damage, fuel and responsibility before the season.

• Check current subsidy or scheme eligibility with the local Agriculture/Horticulture Department; do not assume a subsidy rate or approval.

Routine maintenance

• Follow the manufacturer's service schedule and use the specified oil, filters, belts and lubricants.

• Clean soil, crop residue and chemical deposits after use; dry and store equipment under cover.

• Inspect blades, tines, bearings, tyres, belts, chains, hoses, fasteners and guards before operation.

• Disconnect power, stop the engine and secure the machine before clearing blockages or servicing.

• Drain or protect pumps and lines when needed; clean irrigation filters and check leaks and pressure.

• Calibrate seed drills, planters, fertilizer spreaders and sprayers at the start of each season and when settings change.

Operator safety essentials

• Use trained operators; keep bystanders and children away from operating machinery.

• Wear task-appropriate PPE, including eye, hearing, foot and chemical protection where required.

• Never carry passengers on implements or trailers not designed for passengers.

• Use rollover protection and seat belts where fitted; avoid steep slopes and unsafe towing.

• Do not approach a combine, thresher, baler or PTO shaft while it is moving.

• Follow pesticide labels, re-entry intervals, disposal rules and local recommendations.

Subsidies and hiring

Andhra Pradesh's Agriculture Department has published farm-equipment lists under mechanisation-related programmes, and national programmes have supported equipment access and Custom Hiring Centres. Schemes, eligibility, approved models, application windows and subsidy rates can change. Verify current details with the relevant district Agriculture Office or official state portal before making a purchase.

16. Glossary and reference sources

Quick glossary

Equipment / machine	Size / type	Main field use	Especially relevant for
PTO	Power Take-Off	Tractor output shaft used to power compatible implements	Check speed, guards and shaft compatibility
CHC	Custom Hiring Centre	A service that rents farm machinery to farmers	Useful when expensive machines are needed briefly
HP	Horsepower	A power rating used for engines and tractors	Must match the implement specification
PPE	Personal protective equipment	Protective clothing and equipment for the task	Essential for spraying and machinery work
Zero till	Reduced soil disturbance	Planting without conventional full tillage	Requires suitable equipment and field conditions
Fertigation	Fertilizer + irrigation	Applying compatible nutrients through irrigation water	Use proper filtration and recommendations
Field capacity	Area per time	How much land a machine can cover in a given time	Actual capacity depends on field shape and stops

Reference sources

ICAR — Farm Machinery Principles and Applications
Reference for machinery groups, principles and applications.
https://www.icar.org.in/en/product/46

ICAR — Agricultural Engineering e-books
Covers tillage, sowing, plant protection, harvesting and renewable-energy systems.
https://ebook.icar.org.in/index.php/bookprocess/ugengg_list

ICAR — AICRP on Farm Implements and Machinery
Research and development areas include tillage, sowing, interculture, crop protection and harvesting.
https://icar.gov.in/en/aicrp-farm-implements-and-machinery

Andhra Pradesh Department of Agriculture — farm equipment EOI document
An AP-specific equipment grouping reference; check for newer official documents.
https://apagrisnet.gov.in/images/flash/EOI.pdf

ICAR — Kharif agro-advisories for farmers (2025)
Includes Andhra Pradesh examples of tools for crop and horticulture operations.
https://icar.gov.in/sites/default/files/Circulars/ICAR-En-Kharif-Agro-Advisories-for-Farmers-2025.pdf

Kurnool District, AP — Horticulture Department
Illustrates horticulture and farm-mechanisation support categories in a district context.
https://kurnool.ap.gov.in/horticulture/

Government of India — Sub-Mission on Agricultural Mechanization information
Background on national mechanisation support and access through hiring centres.
https://www.pib.gov.in/PressNoteDetails.aspx?ModuleId=3&NoteId;=159150⟨=1®=3

Image note: The equipment overview illustrations in this guide are generated visual references, not product photographs or certified depictions of specific brands/models. Confirm the exact machine design, specifications and safety features with manufacturers or authorised dealers.

End of guide